

RAYLEIGH–TAYLOR INSTABILITY AND THE EADY PROBLEM

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Mentee

Bismah Zafar

Department of Mathematics

Mount Carmel College, India

Mentor

Dr. Mansi Singh

Mathematical Institute for Machine Learning and Data Science

Katholische Universität Eichstätt–Ingolstadt (KU), Germany

Introduction

This study investigates the Rayleigh-Taylor instability and the Eady problem, which represent key examples of barotropic and baroclinic instabilities, respectively. The primary goal is to analyze these fluid instabilities through numerical and computational methods to gain deeper insights into their dynamics and their implications in atmospheric and fluid dynamics.

In **barotropic systems**, where density is solely a function of pressure, fluid motion remains uniform at different depths. The **Rayleigh-Taylor instability** arises in such systems when a denser fluid is positioned above a lighter one, causing disturbances at the interface. Over time, these disturbances grow and result in the mixing of the two fluids due to the gravity-driven instability.

On the other hand, **baroclinic systems** involve density variations that depend on both pressure and temperature, leading to more intricate flow structures. The **Eady problem**, a fundamental model in the study of mid-latitude atmospheric instabilities, explores how vertical wind shear can induce large-scale disturbances. These disturbances play a significant role in influencing weather patterns and large-scale atmospheric circulation.

Through the numerical modeling of these instabilities, this study seeks to advance our understanding of the underlying mechanisms governing fluid and atmospheric dynamics. Additionally, the findings aim to contribute to broader applications in both fluid mechanics and meteorological science.

Rayleigh-Taylor Instability

1 Derivation of the Rayleigh Equation

Governing Equations

The 2D incompressible Euler equations for a parallel shear flow $\mathbf{U} = (U(y), 0)$ are given by:

$$\frac{\partial u}{\partial t} + U \frac{\partial u}{\partial x} + v \frac{dU}{dy} = -\frac{\partial p}{\partial x}, \quad (1)$$

$$\frac{\partial v}{\partial t} + U \frac{\partial v}{\partial x} = -\frac{\partial p}{\partial y}, \quad (2)$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \quad (3)$$

where $u(x, y, t)$ and $v(x, y, t)$ are the velocity perturbations, and $p(x, y, t)$ is the pressure perturbation.

Streamfunction Representation

Since the flow is incompressible, we introduce a streamfunction $\psi(x, y, t)$ such that:

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}. \quad (4)$$

This automatically satisfies the incompressibility condition.

Vorticity Equation

Taking the curl of the momentum equations eliminates pressure and leads to the vorticity equation:

$$\frac{D\zeta}{Dt} = 0, \quad \text{where } \zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}. \quad (5)$$

Linearizing the vorticity equation around the base flow $U(y)$ gives:

$$\frac{\partial}{\partial t} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) + U \frac{\partial}{\partial x} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) + v \frac{d^2 U}{dy^2} = 0. \quad (6)$$

Using the streamfunction representation:

$$\frac{\partial}{\partial t} (-\nabla^2 \psi) + U \frac{\partial}{\partial x} (-\nabla^2 \psi) + v \frac{d^2 U}{dy^2} = 0. \quad (7)$$

Substituting $v = -\frac{\partial \psi}{\partial x}$, we obtain:

$$\left(\frac{\partial}{\partial t} + U \frac{\partial}{\partial x} \right) \nabla^2 \psi + \frac{d^2 U}{dy^2} \frac{\partial \psi}{\partial x} = 0. \quad (8)$$

Normal Mode Assumption

We assume a normal mode solution of the form:

$$\psi(x, y, t) = \hat{\psi}(y)e^{i(kx - \omega t)}, \quad (9)$$

where k is the wavenumber and ω is the frequency. Substituting this into the vorticity equation gives:

$$(-i\omega + ikU) \left(\frac{d^2 \hat{\psi}}{dy^2} - k^2 \hat{\psi} \right) + ik \frac{d^2 U}{dy^2} \hat{\psi} = 0. \quad (10)$$

Dividing by ik , we obtain the **Rayleigh equation**:

$$(U - c) \left(\frac{d^2 \hat{\psi}}{dy^2} - k^2 \hat{\psi} \right) - \frac{d^2 U}{dy^2} \hat{\psi} = 0, \quad (11)$$

where $c = \frac{\omega}{k}$ is the phase speed of the perturbation.

Boundary Conditions

For physically meaningful solutions, we impose boundary conditions:

- If the flow is confined within walls at $y = \pm h$, then $\hat{\psi} = 0$ at those boundaries.
- If the flow is unbounded (e.g., in a shear layer), then we require the perturbations to decay as $|y| \rightarrow \infty$.

This equation describes the linear stability of an inviscid parallel shear flow. The Rayleigh stability criterion follows from this equation, stating that an inflection point in the base velocity profile ($d^2 U / dy^2 = 0$) is a necessary condition for instability.

2 Numerical Analysis for Different Base Flows

2.1 Inflection Point Analysis

The necessary condition for instability is given by Rayleigh's inflection point theorem, which states that instability can occur only if there exists a point where the second derivative of the velocity profile is zero:

$$U''(y) = 0. \quad (12)$$

We compute $U''(y)$ for each case:

- **Case 1:** $U(y) = 1 - y^2$

$$U'(y) = -2y, \quad U''(y) = -2. \quad (13)$$

Since $U''(y) \neq 0$ for all y , no inflection points exist, meaning this flow is stable.

- **Case 2:** $U(y) = e^{-y^2}$

$$U'(y) = -2ye^{-y^2}, \quad U''(y) = (4y^2 - 2)e^{-y^2}. \quad (14)$$

Setting $U''(y) = 0$ gives:

$$4y^2 - 2 = 0 \Rightarrow y = \pm \frac{1}{\sqrt{2}}. \quad (15)$$

Since inflection points exist, instability is possible.

- **Case 3:** $U(y) = \tanh(y)$

$$U'(y) = \text{sech}^2(y), \quad U''(y) = -2\text{sech}^2(y) \tanh(y). \quad (16)$$

Setting $U''(y) = 0$ gives:

$$-2\text{sech}^2(y) \tanh(y) = 0 \Rightarrow y = 0. \quad (17)$$

Since an inflection point exists, instability is possible.

- **Case 4:** $U(y) = \cos(2y)$

$$U'(y) = -2\sin(2y), \quad U''(y) = -4\cos(2y). \quad (18)$$

Setting $U''(y) = 0$ gives:

$$-4\cos(2y) = 0 \Rightarrow 2y = \frac{\pi}{2} + n\pi, \quad y = \frac{\pi}{4} + n\frac{\pi}{2}, \quad n \in \mathbb{Z}. \quad (19)$$

Since inflection points exist, instability is possible.

2.2 Numerical Solution of the Rayleigh Equation

The Rayleigh equation, governing the perturbation streamfunction $\hat{\psi}(y)$, is given by:

$$(U - c)(\hat{\psi}'' - k^2\hat{\psi}) - U''\hat{\psi} = 0 \quad (20)$$

subject to the boundary conditions:

$$\hat{\psi}(-1) = \hat{\psi}(1) = 0. \quad (21)$$

Given the base flow:

$$U(y) = 1 - y^2, \quad (22)$$

we have:

$$U''(y) = -2. \quad (23)$$

Substituting $U(y)$ and $U''(y)$ into the Rayleigh equation:

$$(1 - y^2 - c) \left(\frac{d^2\hat{\psi}}{dy^2} - k^2\hat{\psi} \right) + 2\hat{\psi} = 0. \quad (24)$$

Rearranging

$$(1 - y^2) (\hat{\psi}'' - k^2\hat{\psi}) + 2\hat{\psi} = c (\hat{\psi}'' - k^2\hat{\psi}) \quad (25)$$

Bringing all terms to one side:

$$(1 - y^2)\hat{\psi}'' - k^2(1 - y^2)\hat{\psi} + 2\hat{\psi} - c(\hat{\psi}'' - k^2\hat{\psi}) = 0 \quad (26)$$

Thus, the equation can be written in the desired form:

$$A\hat{\psi} = cB\hat{\psi} \quad (27)$$

where the differential operators A and B act on $\hat{\psi}$ as follows:

$$A\hat{\psi} = (1 - y^2) \frac{d^2\hat{\psi}}{dy^2} - k^2(1 - y^2)\hat{\psi} + 2\hat{\psi} \quad (28)$$

$$B\hat{\psi} = \frac{d^2\hat{\psi}}{dy^2} - k^2\hat{\psi} \quad (29)$$

To solve this boundary value problem numerically, we discretize the domain $y \in [-1, 1]$ using a finite difference scheme. The second-order derivatives are approximated using central difference formulas, and the resulting linear system is solved for the eigenvalue c and the eigenfunction $\hat{\psi}(y)$. The possibility of having $\text{Im}(c) > 0$ indicates instability.

Discretization of the Rayleigh Equation

We discretize the domain $[-1, 1]$ into N grid points. The grid points are represented as:

$$y_i = -1 + i\Delta y, \quad \text{with} \quad \Delta y = \frac{2}{N-1}, \quad \text{where} \quad i = 0, 1, \dots, N-1.$$

The finite difference approximation for the second derivative is:

$$\frac{d^2 \hat{\psi}}{dy^2} \approx \frac{\hat{\psi}_{i+1} - 2\hat{\psi}_i + \hat{\psi}_{i-1}}{(\Delta y)^2}.$$

This approximation is applied at grid points $i = 1, 2, \dots, N-2$, and at the boundary points $i = 0$ and $i = N-1$, the boundary conditions will be imposed.

Matrix Representation of the Discretized System

Substituting the finite difference approximation into the Rayleigh equation, we get:

$$\begin{aligned} (1 - y_i^2) \frac{1}{(\Delta y)^2} \hat{\psi}_{i+1} - \left(2(1 - y_i^2) \frac{1}{(\Delta y)^2} + k^2(1 - y_i^2) - 2 \right) \hat{\psi}_i + (1 - y_i^2) \frac{1}{(\Delta y)^2} \hat{\psi}_{i-1} \\ = c \left(\frac{1}{(\Delta y)^2} \hat{\psi}_{i+1} - 2\hat{\psi}_i + \hat{\psi}_{i-1} - k^2 \hat{\psi}_i \right) \end{aligned} \quad (30)$$

This equation can be rewritten in matrix form as:

$$A \begin{bmatrix} \hat{\psi}_1 \\ \hat{\psi}_2 \\ \vdots \\ \hat{\psi}_{N-2} \end{bmatrix} = cB \begin{bmatrix} \hat{\psi}_1 \\ \hat{\psi}_2 \\ \vdots \\ \hat{\psi}_{N-2} \end{bmatrix},$$

where the matrices A and B are sparse and tridiagonal. Note that due to the presence of the term $(1 - y_i^2)$, the matrices will not have constant coefficients, but rather these coefficients will depend on the grid point y_i .

Specifically, the matrix A takes the tridiagonal form:

$$A = \begin{bmatrix} a_1 & b_1 & 0 & \cdots & 0 \\ c_2 & a_2 & b_2 & \cdots & 0 \\ 0 & c_3 & a_3 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & b_{N-3} \\ 0 & \cdots & 0 & c_{N-2} & a_{N-2} \end{bmatrix},$$

and similarly for B , where the entries a_i, b_i, c_i depend on the grid point y_i , the base flow profile $U(y_i)$, and the second derivative $U''(y_i)$. These matrices are constructed using central difference approximations of the differential operators A and B , with the boundary values $\hat{\psi}_0 = \hat{\psi}_{N-1} = 0$ enforced.

For clarity, the explicit expressions for the entries of the matrices A and B are given

below. These depend on the base flow $U(y) = 1 - y^2$, the wavenumber k , and the grid spacing Δy :

$$a_i = - \left(\frac{2(1 - y_i^2)}{(\Delta y)^2} + k^2(1 - y_i^2) - 2 \right), \quad b_i = c_i = \frac{(1 - y_i^2)}{(\Delta y)^2}$$

For the matrix B , which arises from the discretization of $\left(\frac{d^2}{dy^2} - k^2\right)\hat{\psi}$, the tridiagonal entries are:

$$b_{ii} = - \left(\frac{2}{(\Delta y)^2} + k^2 \right), \quad b_{i,i-1} = b_{i,i+1} = \frac{1}{(\Delta y)^2}$$

These definitions hold for all interior points $i = 1, \dots, N - 2$, while the boundary conditions $\hat{\psi}_0 = \hat{\psi}_{N-1} = 0$ ensure the matrices are of size $(N - 2) \times (N - 2)$.

Boundary Conditions

The boundary conditions $\hat{\psi}(-1) = 0$ and $\hat{\psi}(1) = 0$ imply that the values $\hat{\psi}_0 = 0$ and $\hat{\psi}_{N-1} = 0$ are enforced at the boundary. These conditions modify the first and last rows of the matrix A , making the corresponding rows zero except for the boundary terms.

Eigenvalue Computation

To find the eigenvalues c , we solve the characteristic equation:

$$\det(A - cB) = 0,$$

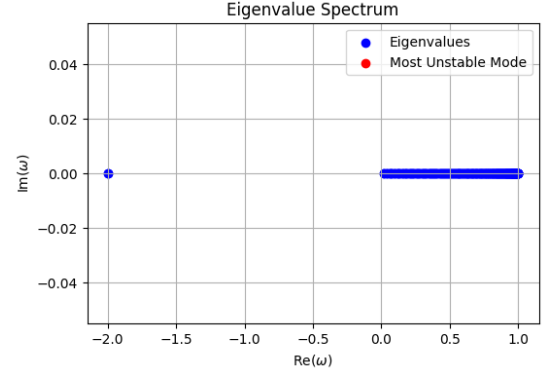
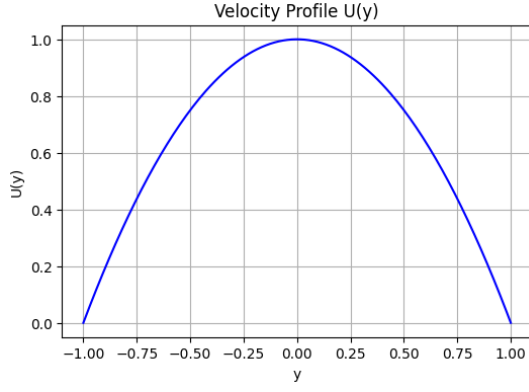
This will allow for the computation of the eigenvalues c , which are the solutions to the Rayleigh equation in the discretized form.

3 Computational Analysis for Different Base Flows

In this section, we analyze the stability of four different base flow velocity profiles by solving the generalized eigenvalue problem. This helps us understand how small disturbances evolve in shear flows and whether they lead to instability. For each case, we compute the eigenvalues to determine stability and visualize the most unstable perturbations. Stream-function and velocity field plots help illustrate how disturbances develop, offering insight into the transition to turbulence.

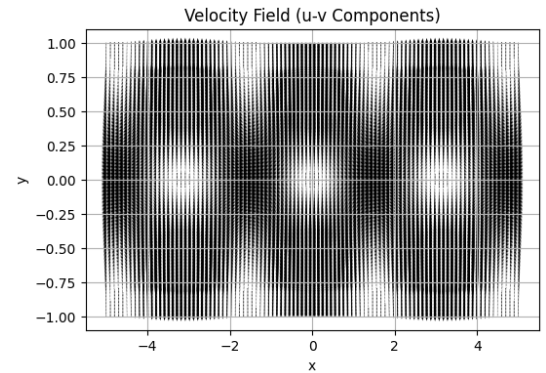
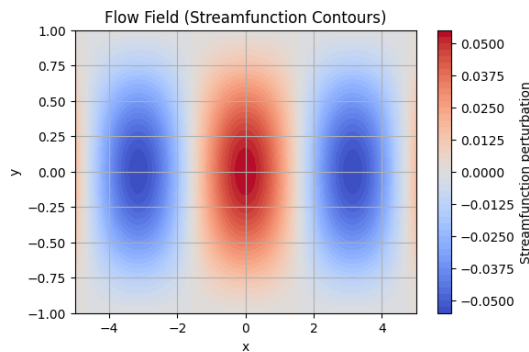
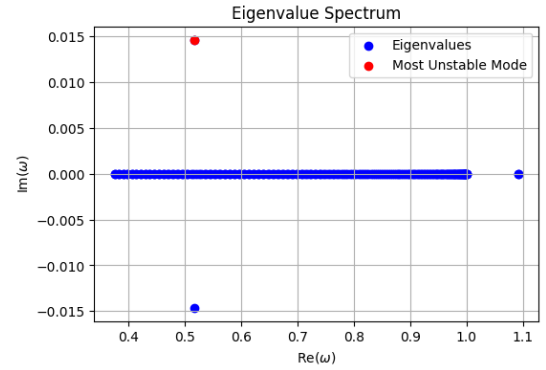
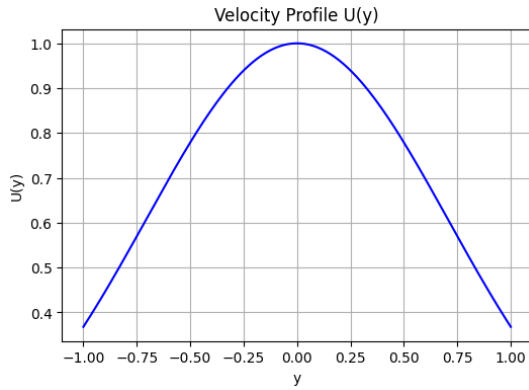
The chosen velocity profiles are:

- **Case 1:** $U(y) = 1 - y^2$ A parabolic velocity profile, characteristic of plane Poiseuille flow, commonly observed in channel flows.

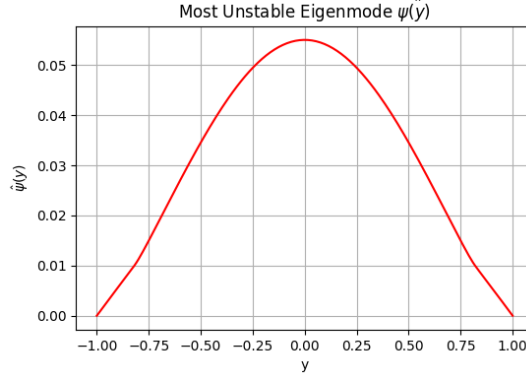


The stability analysis of $U(y) = 1 - y^2$ showed no unstable modes, as all eigenvalues had non-positive imaginary parts, confirming linear stability. Flow perturbations remained bounded, aligning with inviscid stability theory. Further analysis with viscosity or different parameters could offer deeper insights.

- **Case 2:** $U(y) = e^{-y^2}$ A Gaussian profile, often used to model localized shear layers or jet-like velocity distributions.

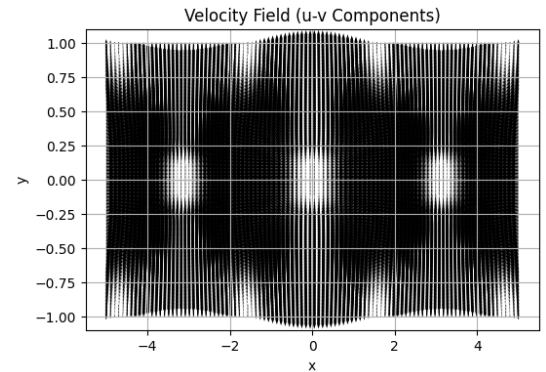
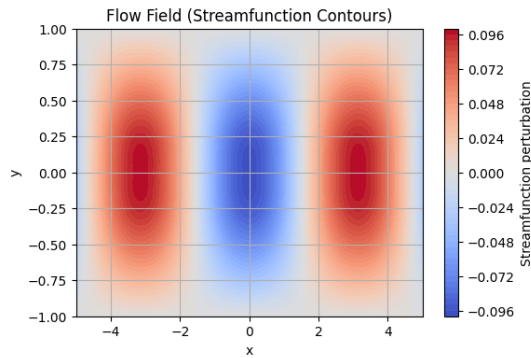
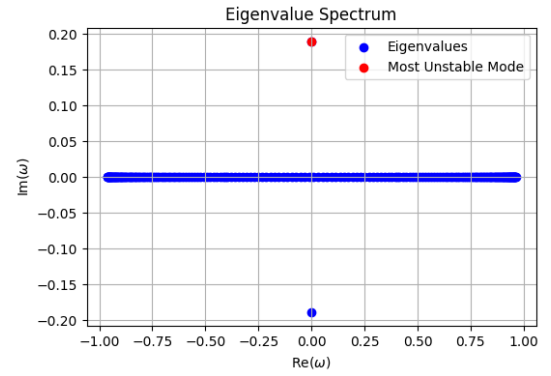
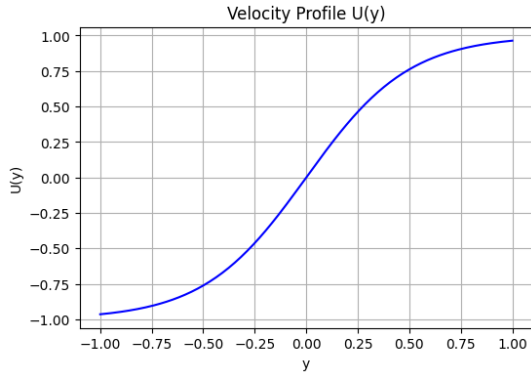


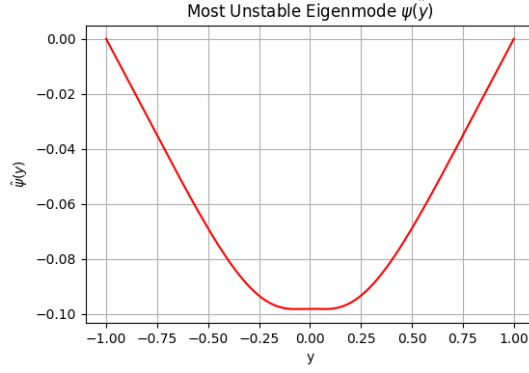
The results illustrate the stability characteristics of the given velocity profile $U(y) = e^{-y^2}$. The eigenvalue spectrum reveals the presence of an unstable mode (red point),



indicating a growing disturbance in the flow. The most unstable eigenmode plot shows the perturbation structure in the y -direction. The flow field contour highlights the streamfunction perturbation, while the quiver plot visualizes the velocity perturbations, illustrating how disturbances evolve spatially. The velocity profile confirms the smooth variation of $U(y)$, influencing the stability behavior. Overall, the analysis suggests an instability in the system due to the identified unstable mode.

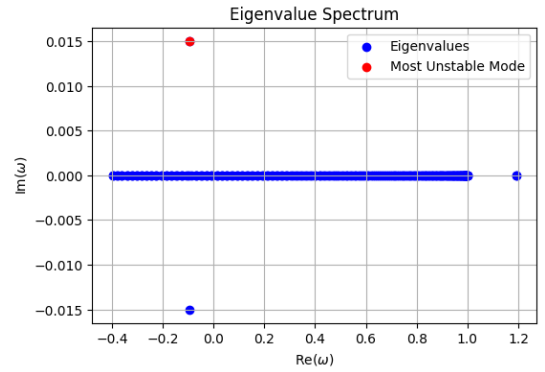
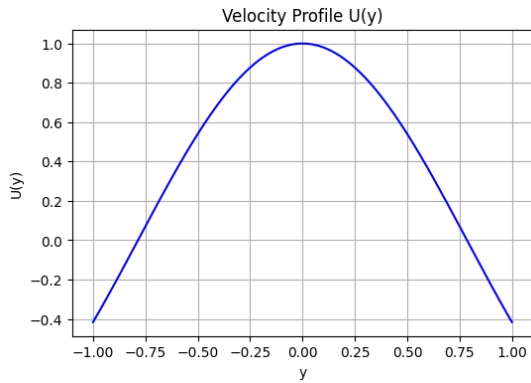
- **Case 3:** $U(y) = \tanh(y/0.5)$ A hyperbolic tangent profile, resembling the velocity distribution in mixing layers and boundary layers.

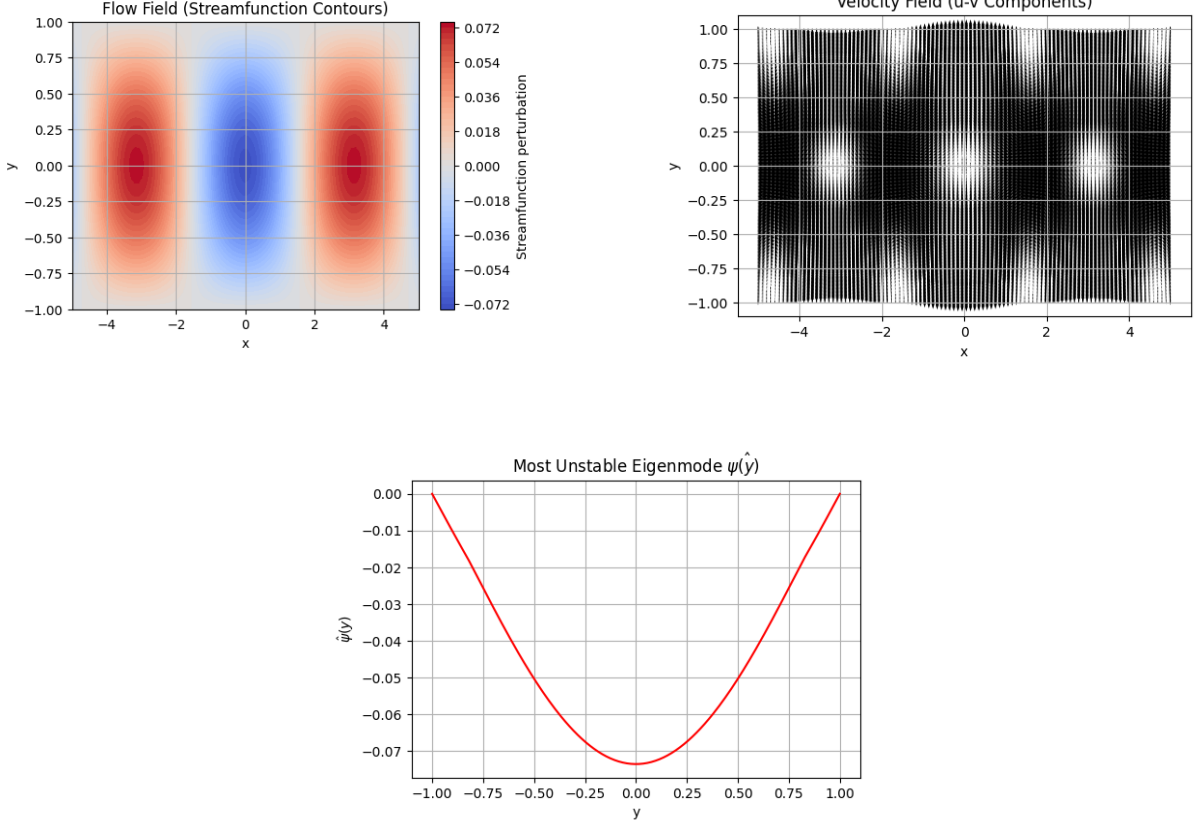




For the velocity profile $U(y) = \tanh(y/0.5)$, the stability analysis reveals the presence of unstable modes, suggesting potential transition to turbulence. The eigenvalue spectrum indicates at least one eigenvalue with a positive imaginary part, confirming instability. The most unstable eigenmode plot highlights the dominant perturbation structure in the y -direction. The streamfunction contour and velocity quiver plots illustrate the evolving disturbance, with alternating flow patterns characteristic of shear-driven instabilities. The velocity profile, resembling a shear layer, is prone to Kelvin-Helmholtz-type instability, which aligns with the observed unstable eigenmodes.

- **Case 4:** $U(y) = \cos(2y)$ A sinusoidal velocity profile, representing periodic shear flows, which can be relevant in stratified or oscillatory flow systems.





For the velocity profile $U(y) = \cos(2y)$, the stability analysis suggests a more complex flow behavior due to the oscillatory nature of the base profile. The eigenvalue spectrum may reveal multiple unstable modes, indicating the potential for instability at different scales. The most unstable eigenmode plot shows how perturbations develop in the y -direction. The streamfunction contour and quiver plots illustrate alternating flow structures, influenced by the periodic shear in $U(y)$. This profile can exhibit instability patterns similar to those found in wave-driven or modulated shear flows, where regions of strong shear contribute to the growth of disturbances.

Remark: It is insightful to compare this with classical base flows such as $U(y) = \tanh(y)$ and $U(y) = \cos(y)$, both of which possess inflection points—a necessary condition for inviscid instability as per Rayleigh’s criterion. However, linear stability analysis under typical conditions reveals no unstable modes for these profiles, indicating that an inflection point alone is not sufficient for instability. The overall structure of the base flow, including the distribution and strength of shear, plays a crucial role. In contrast, profiles like $U(y) = \tanh(y/0.5)$ and $U(y) = \cos(2y)$, with sharper gradients or periodic shear reversal, feature multiple inflection points and alternating shear layers. These characteristics enhance the flow’s receptivity to perturbations, making such profiles more susceptible to instability and capable of supporting richer dynamical behavior.

The Eady Problem

4 Derivation

The Eady problem models baroclinic instability, which describes the growth of disturbances in a stratified fluid, such as the atmosphere or ocean. The key assumptions of the Eady problem are:

- **f-plane:** No variation in the Coriolis parameter ($\beta = 0$).
- **Uniform Stratification:** Constant buoyancy frequency ($N^2 = \text{constant}$).
- **Uniform Shear:** The flow has constant shear with height ($U_0(z) = \Lambda z$), similar to typical atmospheric wind profiles.
- **Rigid Surfaces:** The fluid is contained between two rigid horizontal surfaces, mimicking the ground and tropopause.

The Linearized Problem

With a basic state streamfunction of $\Psi = -\Lambda z y$, the basic state potential vorticity, Q , is

$$Q = \nabla^2 \Psi + \frac{H^2}{L_d^2} \frac{\partial}{\partial z} \left(\frac{\partial \Psi}{\partial z} \right) = 0. \quad (31)$$

The fact that $Q = 0$ makes the Eady problem a special case, albeit an illuminating one. The linearized potential vorticity equation is

$$\left(\frac{\partial}{\partial t} + \Lambda z \frac{\partial}{\partial x} \right) \left(\nabla^2 \psi' + \frac{H^2}{L_d^2} \frac{\partial^2 \psi'}{\partial z^2} \right) = 0. \quad (32)$$

This equation has no x -dependent coefficients and in a periodic channel we may seek solutions of the form $\psi'(x, y, z, t) = \text{Re} \left[\tilde{\psi}(y, z) e^{ik(x-ct)} \right]$, yielding

$$(\Lambda z - c) \left(\frac{\partial^2 \tilde{\psi}}{\partial y^2} + \frac{H^2}{L_d^2} \frac{\partial^2 \tilde{\psi}}{\partial z^2} - k^2 \tilde{\psi} \right) = 0. \quad (33)$$

Boundary Conditions and Solution Form

The Eady problem is governed by the following boundary conditions:

- **Vertical:** The vertical velocity $w = 0$ at both $z = 0$ (surface) and $z = H$ (bottom).
- **Lateral:** The stream function $\Psi = 0$ at $y = \pm \frac{L}{2}$, corresponding to a channel of width L .

To satisfy the lateral boundary conditions, we seek solutions of the form:

$$\psi'(x, y, z, t) = \text{Re} [\Phi(z) \sin(ly) e^{ik(x-ct)}], \quad \text{where} \quad l = \frac{n\pi}{L}, \quad n \in \mathbb{Z}^+. \quad (34)$$

This automatically ensures $\psi' = 0$ at $y = \pm \frac{L}{2}$. The vertical boundary condition $w = 0$ at $z = 0$ and $z = H$ is incorporated through the structure of $\Phi(z)$, and is enforced by the governing equation:

$$\left(\frac{\partial}{\partial t} + \Lambda z \frac{\partial}{\partial x} \right) \frac{\partial \psi'}{\partial z} - \Lambda \frac{\partial \psi'}{\partial x} = 0 \quad \text{at} \quad z = 0, H. \quad (35)$$

Solution

Substituting (34) into (32) gives the interior potential vorticity equation:

$$(\Lambda z - c) \left[\frac{H^2}{L_d^2} \frac{\partial^2 \Phi}{\partial z^2} - (k^2 + l^2) \Phi \right] = 0 \quad (36)$$

Substituting (34) into (35) gives, at $z = 0$ and $z = H$,

$$c \frac{d\Phi}{dz} + \Lambda \Phi = 0, \quad (37)$$

$$(c - \Lambda H) \frac{d\Phi}{dz} + \Lambda \Phi = 0. \quad (38)$$

If $\Lambda z \neq c$ then (36) becomes

$$H^2 \frac{d^2 \Phi}{dz^2} - \mu^2 \Phi = 0, \quad (39)$$

where $\mu^2 = L_d^2(k^2 + l^2)$.

$$\frac{d^2 \Phi}{dz^2} = \frac{\mu^2}{H^2} \Phi \quad (40)$$

Assuming a solution of the form $\Phi(z) = e^{\lambda z}$, we substitute into the equation:

$$\lambda^2 e^{\lambda z} = \frac{\mu^2}{H^2} e^{\lambda z} \quad (41)$$

$$\lambda^2 = \frac{\mu^2}{H^2} \quad (42)$$

$$\lambda = \pm \frac{\mu}{H} \quad (43)$$

The general solution is:

$$\Phi(z) = A e^{\frac{\mu}{H} z} + B e^{-\frac{\mu}{H} z} \quad (44)$$

Using the relations for hyperbolic functions:

$$e^{\frac{\mu}{H}z} = \cosh\left(\frac{\mu}{H}z\right) + \sinh\left(\frac{\mu}{H}z\right) \quad (45)$$

$$e^{-\frac{\mu}{H}z} = \cosh\left(\frac{\mu}{H}z\right) - \sinh\left(\frac{\mu}{H}z\right) \quad (46)$$

The solution can be written as:

$$\Phi(z) = A \cosh(\mu \hat{z}) + B \sinh(\mu \hat{z}) \quad (47)$$

where $\hat{z} = z/H$; thus, μ determines the vertical structure of the solution. The boundary conditions are satisfied if

$$A[\Lambda H] + B[\mu c] = 0, \quad (48)$$

$$A[(c - \Lambda H)\mu \sinh \mu + \Lambda H \cosh \mu] + B[(c - \Lambda H)\mu \cosh \mu + \Lambda H \sinh \mu] = 0. \quad (49)$$

Equations (48) and (49) are two coupled homogeneous equations in the two unknowns A and B . Non-trivial solutions will only exist if the determinant of their coefficients (the terms in square brackets) vanishes, and this leads to

$$c^2 - Uc + U^2(\mu^{-1} \coth \mu - \mu^{-2}) = 0, \quad (50)$$

where $U \equiv \Lambda H$ and $\coth \mu = \cosh \mu / \sinh \mu$.

Solving the above quadratic equation gives the roots as

$$c = \frac{U}{2} \pm \frac{U}{\mu} \left[\left(\frac{\mu}{2} - \coth \frac{\mu}{2} \right) \left(\frac{\mu}{2} - \tanh \frac{\mu}{2} \right) \right]^{1/2}. \quad (51)$$

The waves being proportional to $\exp(-ikct)$, will grow exponentially if c has a strictly positive imaginary part. Since $\mu/2 > \tanh(\mu/2)$ for all μ , for an instability we require that

$$\frac{\mu}{2} < \coth \frac{\mu}{2}, \quad (52)$$

which is satisfied when $\mu < \mu_c$ where $\mu_c = 2.399$. The growth rates of the instabilities themselves are given by the imaginary part of (51), multiplied by the x -wavenumber; that is

$$\sigma = kc_i = k \frac{U}{\mu} \left[\left(\coth \frac{\mu}{2} - \frac{\mu}{2} \right) \left(\frac{\mu}{2} - \tanh \frac{\mu}{2} \right) \right]^{1/2}. \quad (53)$$

5 Computational Analysis

The Eady problem can be solved numerically to obtain the growth rates and characteristics of the most unstable modes for different horizontal wavenumbers and deformation radii. In the following, we show the computational results for various aspects of the Eady problem in nondimensional units.

Atmospheric and Oceanic Parameters

To gain a qualitative sense of the nature of the instability, we select typical parameters for both the atmosphere and ocean:

For the Atmosphere:

$$H \sim 10 \text{ km}, \quad U \sim 10 \text{ m/s}, \quad N \sim 10^{-2} \text{ s}^{-1}$$

From these, we calculate the deformation radius L_d , the scale of maximum instability, and the growth rate:

$$\text{Deformation Radius: } L_d \sim 1000 \text{ km}$$

$$\text{Scale of Maximum Instability: } L_{\max} \sim 4000 \text{ km}$$

$$\text{Growth Rate: } \sigma \sim 0.26 \text{ day}^{-1}$$

For the Ocean:

$$H \sim 1 \text{ km}, \quad U \sim 0.1 \text{ m/s}, \quad N \sim 10^{-2} \text{ s}^{-1}$$

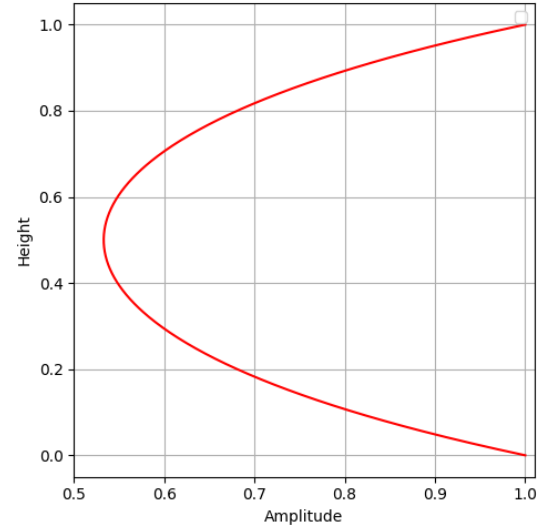
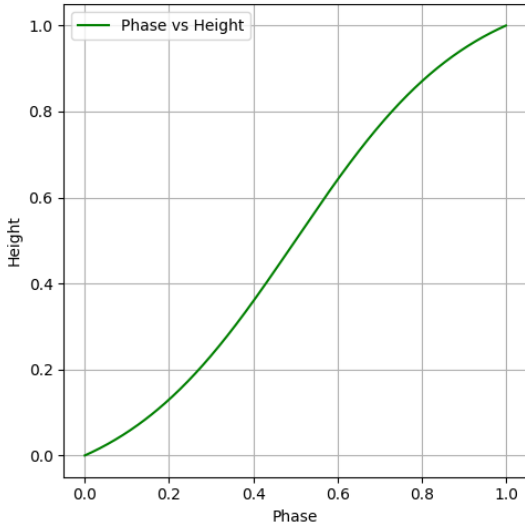
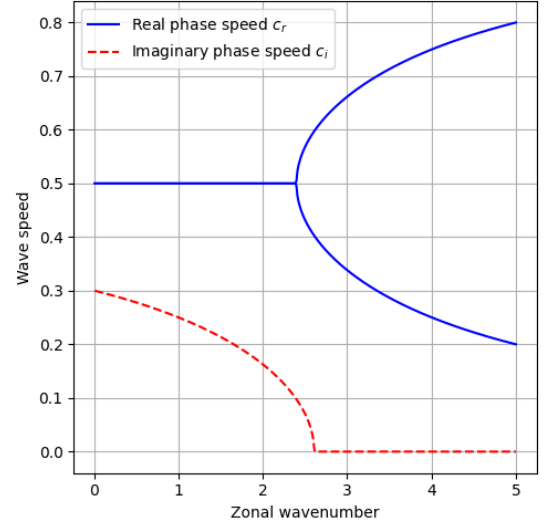
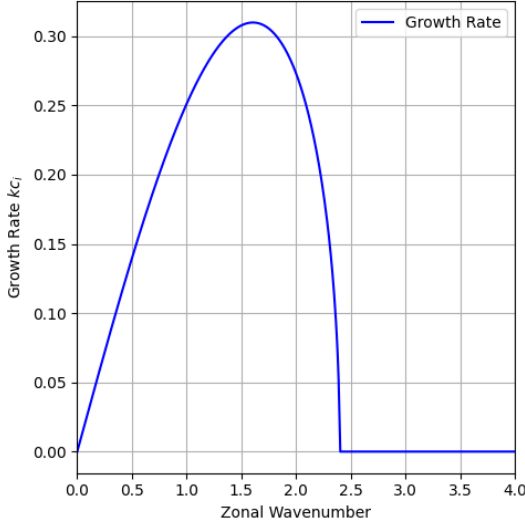
From these parameters, the calculations yield:

$$\text{Deformation Radius: } L_d \sim 100 \text{ km}$$

$$\text{Scale of Maximum Instability: } L_{\max} \sim 400 \text{ km}$$

$$\text{Growth Rate: } \sigma \sim 0.026 \text{ day}^{-1}$$

Solution of the Eady Problem in Nondimensional Units



(a) Growth Rate kc_i as a Function of Scaled Wavenumber μ

The growth rate kc_i is plotted as a function of the scaled wavenumber μ using the formula derived in Equation (53). For the gravest meridional mode, with the assumption that $\Lambda = H = 1$, the growth rate increases with the scaled wavenumber until a maximum is reached. After this, it starts to decrease, demonstrating the dependence of the instability on the wavenumber.

(b) Real (Solid) and Imaginary (Dashed) Wave Speeds as a Function of Horizontal Wavenumber

This panel shows the real and imaginary parts of the wave speed, as functions of the horizontal wavenumber. The real wave speed represents the actual motion of the wave,

while the imaginary component corresponds to the instability rate. As the wavenumber increases, the wave speed approaches a limit, and the imaginary part indicates how the instability grows with increasing horizontal wavenumber.

(c) Phase of the Single Most Unstable Mode as a Function of Height

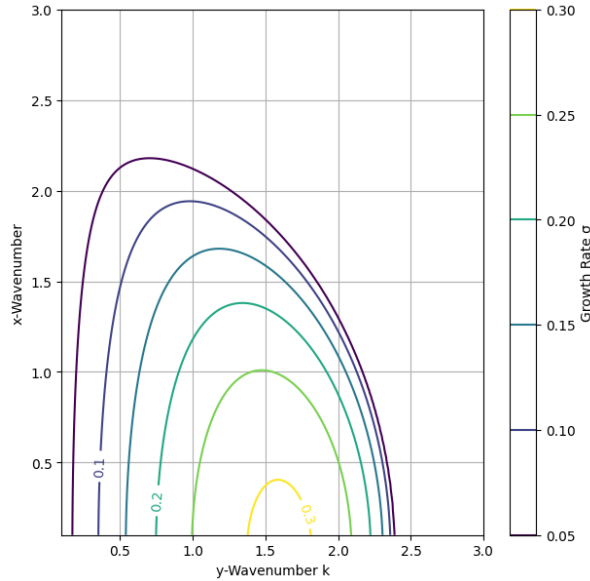
The phase of the most unstable mode is plotted against height to show how the wave structure varies with depth in the fluid. The phase transition is essential in understanding the vertical propagation of disturbances and their role in baroclinic instability.

(d) Amplitude of the Single Most Unstable Mode as a Function of Height

The amplitude plot of the most unstable mode as a function of height illustrates how the intensity of the wave changes with altitude. Typically, the amplitude is largest near the surface and decays with height, reflecting the vertical structure of the instability.

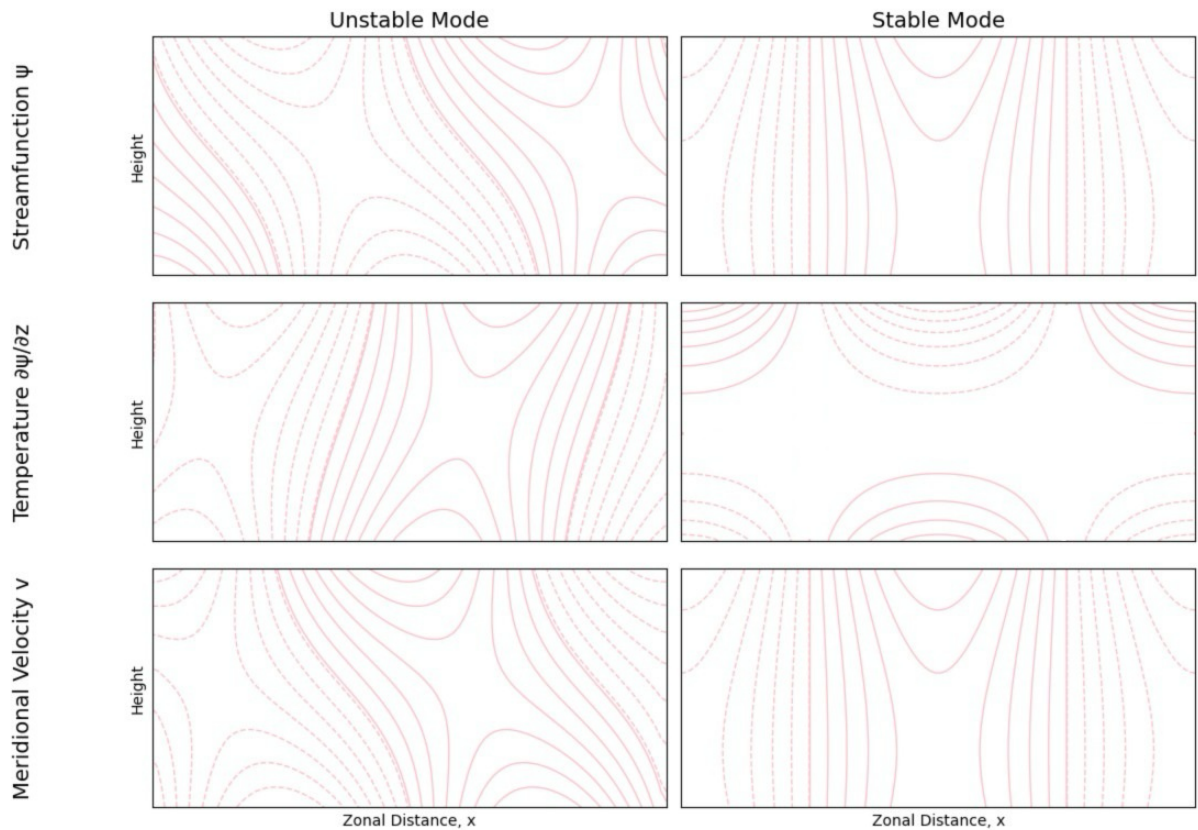
To obtain dimensional values from these results, one can multiply the growth rate by $\frac{\Delta H}{L_d}$ and the wavenumber by $\frac{1}{L_d}$, where L_d is the deformation radius.

Contours of the Growth Rate σ in the $k - l$ Plane



This figure presents the contours of the growth rate σ in the $k - l$ plane, derived from Equation (53). The growth rate peaks near the deformation scale, and for any given zonal wavenumber, the most unstable wave number corresponds to the gravest meridional scale. The figure visually captures the relationship between the horizontal wavenumber k and the meridional wavenumber l in determining the instability characteristics.

Some Results based on the Most Unstable Eady Mode



The plots illustrate the transition from a most unstable Eady wave (left column) to a wave beyond the short-wave cutoff (right column), highlighting the effects of baroclinic instability.

- In the unstable wave, the streamfunction exhibits a clear westward tilt with height, indicating a strong baroclinic instability. The temperature gradient shows a clear poleward heat transport, which is characteristic of this instability. The meridional velocity is stronger in the poleward direction, consistent with the dynamics of mid-latitude cyclones.
- As the wave moves to the right column (beyond the short-wave cutoff), the streamfunction becomes more vertically aligned, losing the westward tilt. This indicates a reduction in the instability's intensity. The temperature gradients become less pronounced, and the meridional velocity weakens, reflecting a more stable flow with little to no energy growth.

In summary, the left column demonstrates the energetic growth of a baroclinically unstable wave, while the right column shows a transition to a neutral or stable wave, where instability and wave energy are significantly diminished.

6 Conclusion

This work has provided a comprehensive investigation into the roles of barotropic and baroclinic instabilities in atmospheric and fluid dynamics. By studying the governing equations for the Rayleigh-Taylor instability and the Eady problem, valuable insights were gained into the mechanisms driving these instabilities.

The development and implementation of computational models enabled the simulation of both the Rayleigh-Taylor instability and the Eady problem, using numerical methods to examine stability conditions. The computational analysis effectively compared theoretical predictions with numerical solutions, providing a deeper understanding of the growth rates, wavenumbers, and structural features of the unstable modes. This comparison demonstrated the accuracy and reliability of numerical models in capturing the essential dynamics of these complex phenomena.

The expected outcomes of the study were achieved, shedding light on the behavior of these instabilities and their impact on fluid dynamics. The results emphasized the importance of computational techniques in modeling and analyzing such instabilities, especially when analytical solutions are difficult to obtain.

Looking ahead, there are numerous opportunities to expand this research. Future work could explore matrix-free eigenvalue problems for nonlinear dynamical systems, incorporating methods such as Dynamic Mode Decomposition (DMD) for spectral analysis. Additionally, data-driven techniques like Singular Value Decomposition (SVD) could be employed for the linearization of complex dynamical systems, which would allow for a more detailed study of the stability of higher-dimensional systems. These directions could lead to a more advanced understanding of nonlinear instabilities and their applications in atmospheric and fluid dynamics.

In summary, the work carried out has laid a solid foundation for future investigations into Rayleigh-Taylor and Eady instabilities, providing both theoretical insights and computational techniques that will contribute to further advancements in the field of fluid mechanics and dynamical systems.

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